FuelCast: 30 -Day Predictive Analytics Platform for Local Fuel Prices

A Web and Mobile UI/UX project

Presented to the Faculty of the

Multimedia Arts Department, iACADEMY Cebu

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RESEARCHING

Project Overview

FuelCast is a tool that helps forecast fuel prices in the Philippines one month ahead. It gathers information from different sources such as international oil and gas prices, foreign exchange rates, and local economic growth. This data is stored in a database and processed using machine learning models to predict if prices will rise, fall, or remain stable.

The system was created to support businesses and drivers who struggle with sudden fuel price changes. By giving early forecasts, the project allows transport companies and logistics managers to plan their budgets, set delivery fees, and manage expenses more effectively.

Problem Statement

Fuel prices in the Philippines are unstable because of global market changes and local supply issues. Businesses and drivers often face problems when prices suddenly increase, making it hard to plan monthly expenses. Current tools only show real-time prices but do not give forecasts for the coming weeks.

This system solves the problem by providing a one-month forecast. It helps companies prepare for possible price changes, manage contracts better, and control operating costs. By knowing the trend early, businesses can make smarter financial decisions and avoid unexpected losses.

Target Audience

The primary target users of the proposed system are logistics managers and transport companies in the Philippines who rely heavily on fuel for their daily operations. These users need a reliable way to forecast gasoline prices one month ahead so they can plan delivery fees, manage budgets, and avoid sudden increases in operating costs. The system helps them prepare for future expenses and make smarter financial decisions.

The secondary target users are business owners. Business owners require accurate forecasts to negotiate contracts, stabilize monthly expenses, and reduce risks from fuel price changes. The proposed system aims to improve financial stability and provide clear insights for both private companies and public institutions.

The third target users are student drivers and individual vehicle owners. These users manage tight personal or allowance-based budgets and face daily financial pressure from sudden gasoline price spikes. By having access to a simple one-month forecast, student drivers and everyday vehicle owners can better plan their monthly travel schedules, budget for campus commutes, and manage their personal expenses effectively.

Competitor Analysis

Features	Accessibility	Readability	Clear Navigation	Visual Design	Functionality	Overall User Experience
Fuel Watchers	✔	✔	✔			
Smart Gasph	✔	✔			✔	
Getodigo		✔	✔	✔	✔	
FuelCast	✔	✔	✔	✔	✔	✔

Table 1: **Comparative Matrix**

A competitor analysis was conducted to evaluate existing systems and identify design practices that can improve FuelCast.

One of the competitors studied was FuelWatchers, a community-driven platform that tracks real-time station prices across the Philippines. This app is highly accessible and easy to navigate for monitoring everyday gas station prices. Its main strength lies in community engagement and localized price tracking. However, it relies heavily on user submissions, which can lead to outdated information, and it lacks automated predictive analytics. The proposed system can adopt its clean navigation and user-friendly interface while providing automated data collection and forward-looking forecasts.

Another competitor is SmartGas PH, a local price monitoring platform that provides daily fuel advisories and alerts. It offers high readability and functionality for users seeking immediate notification of weekly price adjustments. Its weakness is that it only focuses on short-term news announcements rather than medium-term trends. As a result, it cannot help fleet managers or businesses with long-term budget planning. The proposed system can incorporate its clear alert mechanisms while expanding functionality to deliver 30-day price trend predictions.

Lastly, GetOdigo is a fleet management platform that offers strong visual design and enterprise-level tracking for logistics companies. Its strength lies in comprehensive expense management and operational efficiency tools. However, its fuel tracking features are purely historical and reactive, meaning it records past spending without forecasting future fuel price shifts. FuelCast can adopt its professional dashboard layout and fleet analytics tools while adding a predictive engine to give businesses a true one-month financial planning window.

The FuelCast stands out because it combines the strengths of all competitors—accessibility, clear navigation, modern visual design, and real-time tracking—while adding automated 30-day machine learning forecasts. This unique approach ensures logistics managers, business owners, and everyday drivers in the Philippines receive actionable insights to plan budgets and improve financial stability.

User Personas

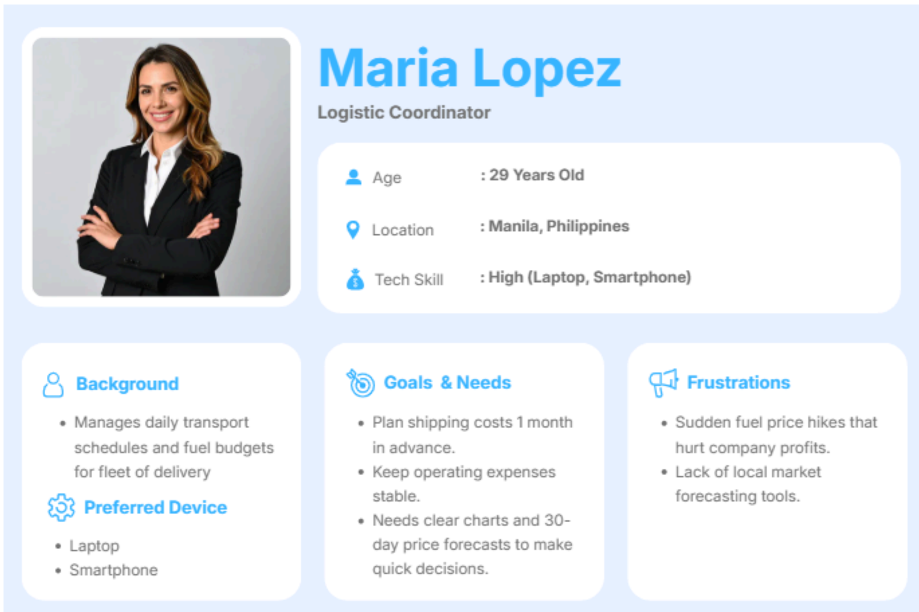


Figure 1: **Persona One - Maria Lopez**

Maria Lopez is a 29-year-old logistics coordinator in Manila. She manages delivery schedules and fuel budgets for a fleet of vans. Her main goal is to plan shipping costs a month early and keep business expenses stable. She needs a simple system with 1-month fuel price forecasts and clear charts to make fast decisions. Maria gets frustrated by sudden gas price hikes that hurt company profits. She is very tech-savvy and prefers using her laptop and smartphone to check price trends.

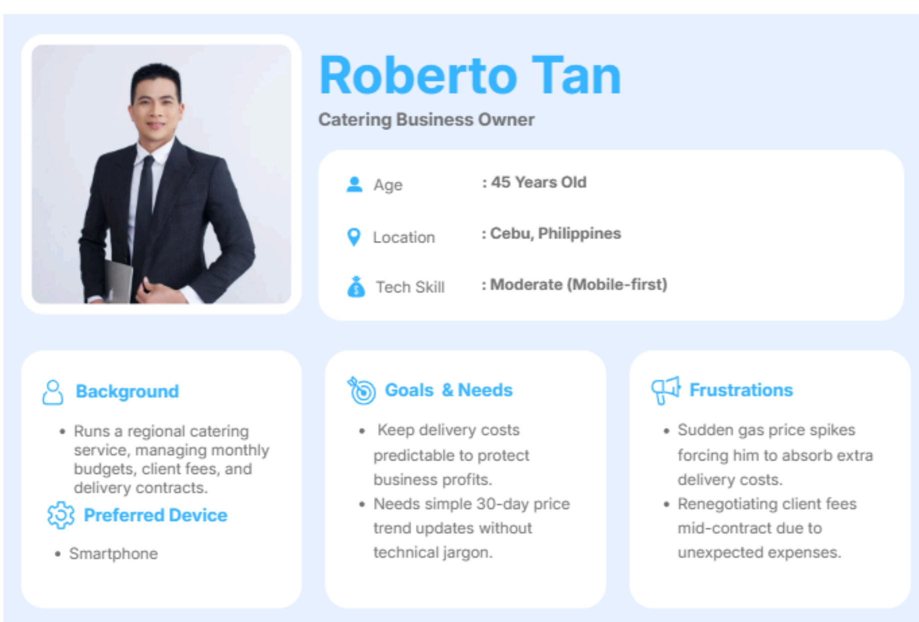


Figure 2: **Persona Two - Roberto Tan**

Roberto Tan is a 45-year-old catering business owner in Cebu. He sets monthly budgets, manages client delivery fees, and plans long-term contracts. His main goal is to keep his shipping costs predictable so he does not lose profit. He needs a simple tool that gives clear 30-day price trend updates without confusing math. Roberto gets frustrated when sudden fuel price spikes force him to pay extra shipping fees out of his own pocket. He uses his smartphone daily for business and prefers getting quick visual reports on his phone.Unexpected fuel cost spikes have also forced him to renegotiate client fees mid-contract, straining relationships with long-term customers he depends on for steady business.

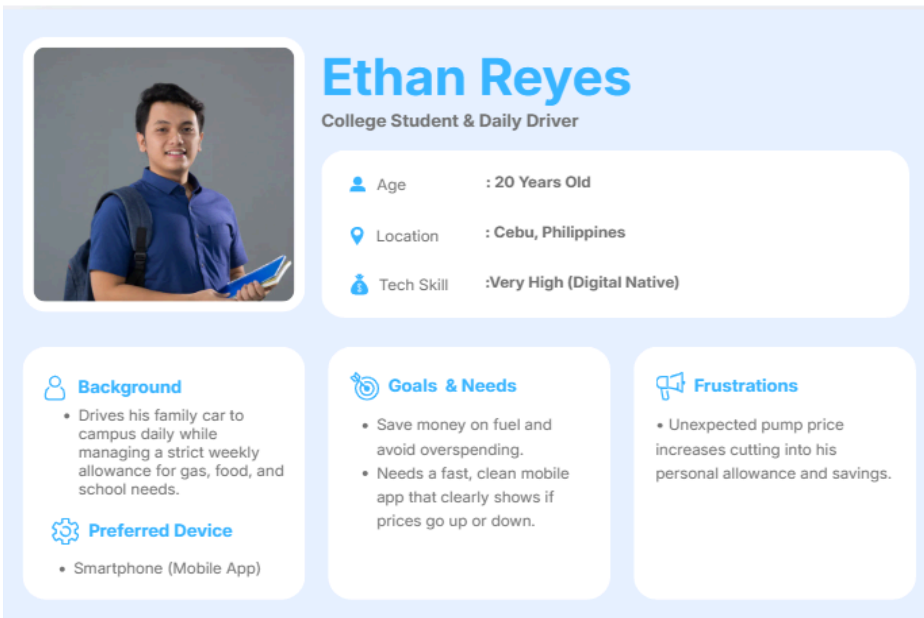


Figure 3: **Persona Three - Ethan Reyes**

Ethan Reyes is a 20-year-old college student in Don Bosco Technical College- Cebu who drives to school every day. He lives on a strict weekly allowance for fuel, food, and school needs. His main goal is to save money on gas and avoid overspending during busy school weeks. He needs a clean mobile friendly web app that shows if gas prices will go up or down over the next month. Ethan gets frustrated when unexpected price hikes cut into his personal savings. He is very tech-savvy and prefers checking fast, easy visual alerts on his smartphone.

User Research Summary

The research aimed to understand how individual drivers, logistics coordinators, and small business owners currently manage fuel budgeting and identify the challenges they face when fuel prices change unexpectedly. To gather relevant data, an online survey was conducted with 11 respondents across the Visayas region of the Philippines. Among the respondents, 64% were students or daily individual drivers, 18% were logistics/transport coordinators, and 18% were business owners, with the majority (64%) falling within the 20–29 age range.

The survey results revealed that 91% of respondents cited sudden, unpredictable price spikes as their biggest fuel-related difficulty, and 91% currently track price changes only through news and social media, with no dedicated tool. Additionally, 82% expressed dissatisfaction with existing tools that show only current or historical prices rather than forecasts, 91% rated a 30-day price forecast as extremely or very useful, and 64% said a one-week advance warning was the ideal lead time to adjust their routine. Furthermore, 82% preferred a clean, color-coded visual dashboard over text-only or table-based designs, and 73% favored simple tab-based navigation. These findings suggest a strong demand for a predictive, visually simple digital tool that current fuel-tracking solutions do not provide.

Based on the research results, several design implications were identified. FuelCast will prioritize a one-week price-increase alert as its core, action-driving feature, supported by the full 30-day forecast for broader planning. The interface will adopt a color-coded, tab-based dashboard design to match user preference for quick, visual decision-making, with drill-down tables and short plain-language explanations available for users who want more detail. A trip/route fuel cost calculator will be incorporated as a secondary feature to address the most-requested addition beyond forecasting itself.

Overall, the research confirms that fuel users in this market would benefit from a predictive, easy-to-navigate forecasting platform that replaces reactive, news-based tracking with proactive alerts, resulting in better-protected budgets and more confident fuel purchasing decisions.

PHASE 2

PLANNING

Information Architecture

Information Architecture (IA) refers to the organization and structure of information within a digital product. It serves as the blueprint of the system by defining how content, features, and pages are arranged to help users easily find information and complete tasks.

For MotoBook, the information architecture was designed based on the findings gathered during the research phase. Since the primary task of users is booking motorcycle maintenance services, the system prioritizes quick access to service listings, appointment scheduling, and booking management.

The platform consists of two primary user groups: Motorcycle Owners and Service Centers. Each user type is provided with features relevant to their needs.

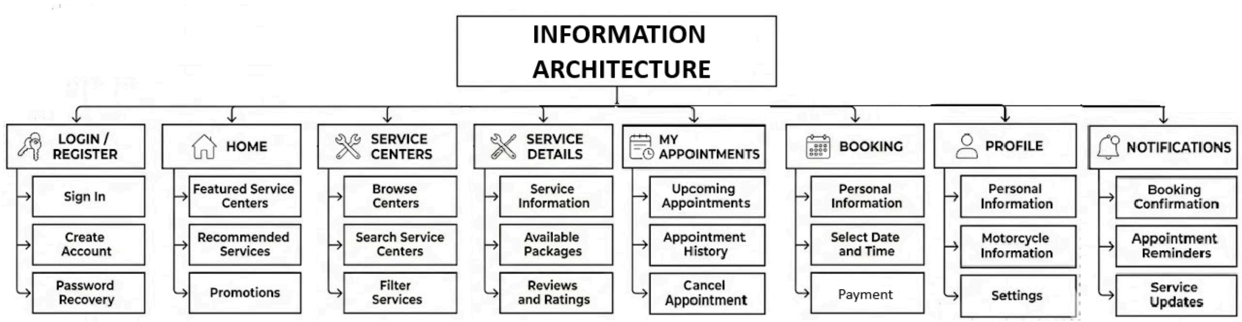


Figure 3: **<u>Information Architecture of MotoBook</u>**

The structure was organized using a hierarchical approach where users can move from general categories to specific actions. This minimizes navigation complexity and supports efficient task completion.

Screen List

The following screens were identified as necessary to support the core functionalities of the MotoBook platform.

Screen Name	**Purpose**
Splash Screen	Displays application branding during startup
Onboarding Screen	Introduces platform features to new users
Login Screen	Allows users to access their accounts
Registration Screen	Enables account creation
Forgot Password Screen	Allows password recovery
Home Screen	Displays featured services and quick actions
Service Centers Screen	Lists available service centers
Search Results Screen	Displays filtered service center results
Service Details Screen	Shows information about a selected service center
Booking Screen	Allows users to schedule appointments
Booking Confirmation Screen	Confirms successful booking
My Appointments Screen	Displays upcoming and previous bookings
Notifications Screen	Shows reminders and updates
Profile Screen	Displays user information and settings
Edit Profile Screen	Allows profile modification

Table 2: **Screen List for MotoBook**

The identified screens were selected to ensure that users can complete their primary tasks efficiently while minimizing unnecessary navigation steps.

Low-Fidelity Wireframes

Low-fidelity wireframes were created to establish the initial structure and layout of the application before applying visual design elements. These wireframes focus on content placement, navigation flow, and feature organization rather than colors, typography, or branding.

The wireframes were designed using grayscale placeholders to allow rapid iteration and evaluation of usability concerns. Key screens developed during this stage include:

Home Screen Wireframe:

[Insert Home Screen Wireframe]

Figure 4: **Home Screen Low-Fidelity Wireframe**

The Home Screen serves as the primary landing page of the MotoBook platform. It provides users with quick access to essential features such as searching for service centers, viewing recommended services, and accessing upcoming appointments. The layout is designed to prioritize frequently used actions while maintaining a clear visual hierarchy. This screen helps users efficiently begin their service booking journey and navigate to other sections of the application.

Service Details Screen Wireframe:

[Insert Service Details Wireframe]

Figure 5: **Service Details Low-Fidelity Wireframe**

The Service Details Screen presents comprehensive information about a selected motorcycle service center. It includes details such as available maintenance packages, service descriptions, operating hours, customer reviews, and pricing information. The screen is designed to help users make informed decisions before scheduling an appointment. By organizing information into clearly defined sections, users can quickly compare services and proceed with the booking process confidently.

Booking Screen Wireframe:

[Insert Booking Screen Wireframe]

Figure 6: **Booking Screen Low-Fidelity Wireframe**

The Booking Screen allows users to schedule motorcycle maintenance appointments with their preferred service center. It includes options for selecting a service package, choosing an available date and time, and reviewing booking details before confirmation. The layout follows a step-by-step structure to reduce confusion and guide users through the scheduling process. This screen aims to simplify appointment booking while minimizing user effort and potential errors.

[and so on]

User Flow

User flow diagrams were created to visualize how users navigate through the platform to complete specific tasks. These flows help identify potential usability issues and ensure that users can accomplish goals with minimal friction.

[Insert User Flow Diagram]

Figure 7: **Appointment Booking User Flow**

The Appointment Booking User Flow illustrates the sequence of steps a motorcycle owner follows to successfully schedule a maintenance service through the MotoBook platform. The process begins when the user logs into the application and accesses the Home Screen. From there, the user can browse or search for available service centers, review service details, and select a preferred maintenance package. After choosing an available date and time slot, the user proceeds to confirm the booking. Once the transaction is completed, the system displays a booking confirmation and stores the appointment under the user's appointment history.

The flow was designed to provide a clear and straightforward path toward completing the primary task of the application. By minimizing unnecessary steps and presenting information in a logical sequence, the platform reduces user effort, improves navigation efficiency, and enhances the overall user experience. Additionally, confirmation messages and appointment tracking features help ensure that users remain informed throughout the booking process.

(You may break down your user flow to different tasks e.g. flow for registration, flow for booking, etc.)

Accessibility Considerations

Accessibility considerations were incorporated during the planning phase to ensure that MotoBook can be used effectively by a wide range of users, including those with visual, motor, and cognitive limitations.

Visual Accessibility:

- Text sizes will maintain a minimum readable size of 16px.
- High contrast ratios will be used between text and background colors.
- Icons will be paired with text labels whenever possible.
- Important actions will use clear visual emphasis.

Motor Accessibility:

- Interactive buttons will follow recommended touch target sizes.
- Adequate spacing will be provided between clickable elements.
- Essential actions will require minimal gestures.

Cognitive Accessibility:

- Consistent navigation patterns will be maintained throughout the platform.
- Clear and concise labels will be used.
- Booking processes will be divided into manageable steps.
- Confirmation messages will provide immediate feedback after actions.

Mobile Accessibility:

- Layouts will be responsive across different screen sizes.
- Forms will support mobile-friendly input fields.
- Error messages will clearly explain how users can resolve issues.

By integrating accessibility principles during the planning stage, the proposed platform aims to provide a more inclusive and user-friendly experience for all motorcycle owners and service center personnel.

APPENDIX A

Survey Questionnaire

Objective: To understand the current maintenance booking practices of motorcycle owners, identify common challenges encountered during service scheduling, and determine the features that can improve the convenience, efficiency, and overall user experience of motorcycle maintenance appointments.

Questions:

Personal Budgeting & Allowance Actions

When gas prices spike unexpectedly, how does it affect your personal spending? *

- I have to cut back on food, school funds, or daily needs
- It reduces my weekly savings
- I reduce my driving trips
- It has little effect on me

If FuelCast alerts you that gas prices will rise by PHP 2.00 next week, what will you do? *

- Refuel my vehicle full before the price hike happens
- Plan fewer trips to save gas
- Adjust my daily budget to cover the cost
- Take no action

If FuelCast alert you that gas prices will drop next week, what will you do? *

- Buy only fuel for current needs and wait for the price drop
- Stick to my normal refueling schedule
- Take no action

How far in advance do you need a price warning to adjust your personal routine? *

- 2 to 3 days in advance
- 1 week in advance
- 2 weeks in advance
- 30 days in advance

User Demographic & Profile

What best describes your main role when purchasing fuel or managing transport? *

- Logistics Manager / Transport Coordinator e.g., (Lalamove, J&T Express, or Ninja Van)
- Business Owner
- Student / Daily Individual Driver

What is your age? *

- Below 20 years old
- 20 – 29 years old
- 30 – 44 years old
- 45 years old and above

Where are you located in the Philippines? *

- Luzon
- Visayas
- Mindanao

Which device do you mostly use for work, managing budgets, or tracking prices? *

- Smartphone
- Laptop / Desktop
- Both Smartphone and Laptop
- Other:

How comfortable are you with using smartphone applications and digital tools? *

- Low (I prefer manual paper records)
- Moderate (I use basic daily smartphone apps)
- High (I easily learn new apps and desktop software)
- Very High (I use digital devices for almost everything)

Current Fuel Management & Financial Pain Points

How often do you unexpected fuel price increases harm you personal or business budget? *

- [ ] Always
- [ ] Often
- [x] Sometimes
- [ ] Rarely
- [ ] Never

What is your biggest difficulty when dealing with fuel price changes? *

- [x] Sudden price spikes that cut into my profits or personal savings
- [ ] Setting delivery fees or monthly travel budgets in advance
- [ ] Lack of simple tools that forecast future price shifts
- [ ] Other: ____

How do you currently track or prepare for weekly fuel price changes? *

- [x] Watching news announcements on television or social media
- [ ] Checking real-time gas station monitoring apps
- [ ] I do not prepare; I only react after prices change
- [ ] Other: ____

How far in advance do you attempt to plan your delivery costs or personal transport budgets? *

- [x] Weekly
- [ ] 1 Month in advance
- [ ] Multi-month or quarterly
- [ ] I do not plan ahead

How satisfied are you with existing fuel tools that only display current or historical prices? *

- [ ] Very satisfied
- [ ] Satisfied
- [x] Neutral
- [ ] Dissatisfied
- [ ] Very dissatisfied

Existing Market Tools

Which of the following fuel tracking platforms have you used before? *(Select all that apply)* *

- [ ] Fuel Watchers
- [ ] SmartGas PH
- [ ] GetOdigo
- [x] News or Social Media pages
- [ ] None

Which design aspect is most important to you when choosing a tracking app? *

- [x] Clear visual design and modern layout
- [ ] Simple, easy to use menu navigation
- [ ] Simple language without technical jargon

FuelCast Core Feature Assessment

How useful would a tool that forecasts fuel prices month (30 days) ahead be for you? *

- [ ] Extremely Useful
- [ ] Very Useful
- [ ] Somewhat Useful
- [ ] Not Useful

FuelCast predicts whether will rise, fall, or stay stable. Which information helps you the most?

- [ ] Early warnings about price increases
- [ ] Monthly trend stabilty
- [ ] Other: ____

How would you like the 30-day forecast to be presented on your screen *

- [ ] Clear visual charts and simple line graphs
- [ ] Simple color Indicators (Red for Increase, Green for Decrease)
- [ ] Detailed data tables with numbers and percentages

FuelCast uses automated systems combining shipping, market, and economic data to make predictions. Does this build your trust in the system? *

- [ ] Yes
- [ ] No
- [ ] Neutral

FuelCast Core Feature Assessment

How useful would a tool that forecasts fuel prices month (30 days) ahead be for you? *

- ○ Extremely Useful
- ◉ Very Useful
- ○ Somewhat Useful
- ○ Not Useful

FuelCast predicts whether will rise, fall, or stay stable. Which information helps you the most?

- ◉ Early warnings about price increases
- ○ Monthly trend stabilty
- ○ Other: ______

Clear selection

How would you like the 30-day forecast to be presented on your screen *

- ☑ Clear visual charts and simple line graphs
- ☐ Simple color Indicators (Red for Increase, Green for Decrease)
- ☐ Detailed data tables with numbers and percentages

FuelCast uses automated systems combining shipping, market, and economic data to make predictions. Does this build your trust in the system? *

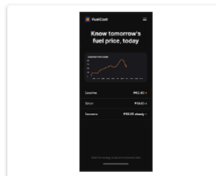
- ◉ Yes
- ○ No
- ○ Neutral

Usability & User Interface Preferences

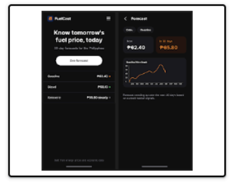
Which visual style makes it easier for you to make quick decisions? *

- ◉ Clean visual dashboard with color-coded alerts
- ○ Minimalist design with text only
- ○ Standard table or list view
- ○ Other: ______

How do you prefer to navigate through price trends inside the web application? *



- ○ Simple tab views



- ◉ Simple navigation buttons with visual icons

How should FuelCast explain complex forecast predictions to you? *

- ◉ Show simple trend arrows and plain text descriptions without technical math
- ○ Provide short summary notes explaining the main reasons for the price change
- ○ Other: ______

Operational & Contract Decisions

If fuel prices suddenly increase, what actions do you usually take? *

- ◉ Absorb the extra costs out of pocket, which hurts profits or personal budgets
- ○ Attempt to renegotiate delivery rates mid-contract
- ○ Reduce non-essential travel or adjust shipping schedules
- ○ Fill up vehicle tanks immediately before the price hike

Which additional expense tool would be most helpful inside FuelCast *

- ○ Simple trip our route fuel cost calculator
- ○ Fleet fuel consumption and expense log
- ◉ Local station price finder
- ○ None

How would predictable fuel costs affect your operational overall? *

- ◉ Protects profit margins and personal allowance
- ○ Saves time spent adjusting monthly spending plans
- ○ Builds trust with delivery clients through stable pricing
- ○ Has minimal impact

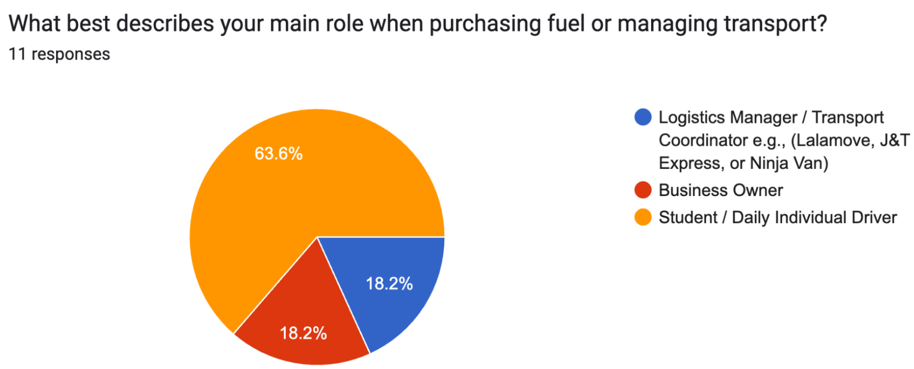
How much time do you currently spend adjusting budgets due to changing gas prices *

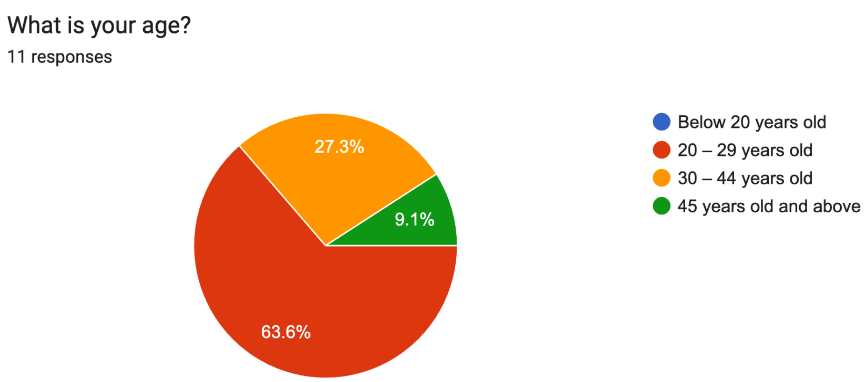
- ○ Less than 1 hour per month
- ◉ 1 to 5 hours per month
- ○ More than 5 hours per month
- ○ I do not track budget change

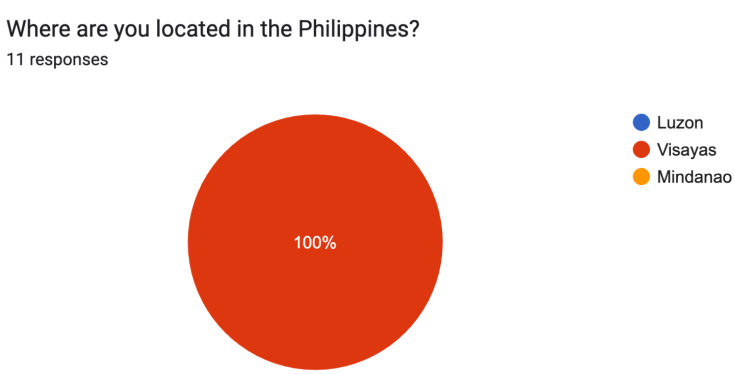
APPENDIX B

Survey Responses

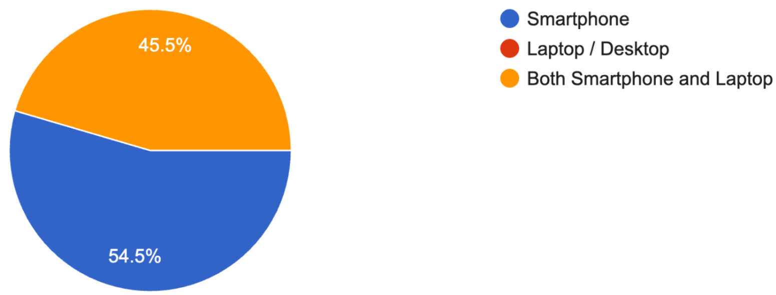
The following results and findings are based on the data acquired from the respondents through surveys.



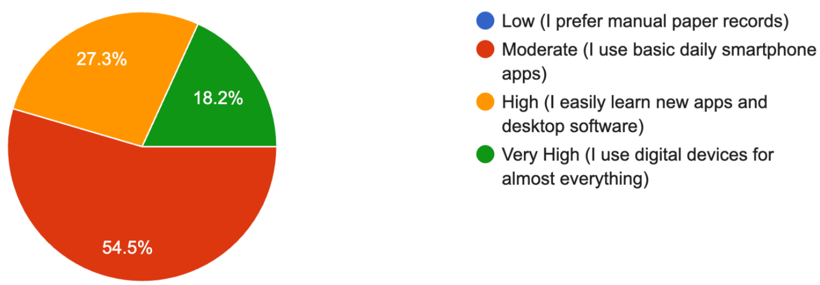




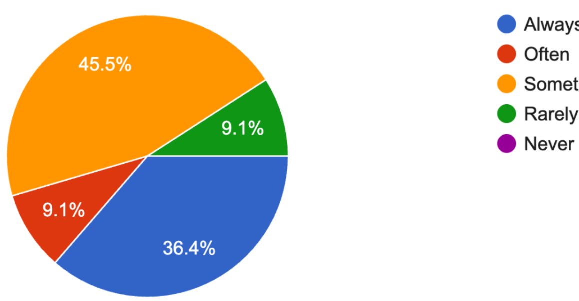
Which device do you mostly use for work, managing budgets, or tracking prices?
11 responses



How comfortable are you with using smartphone applications and digital tools?
11 responses



How often do you unexpected fuel price increases harm you personal or business budget?
11 responses



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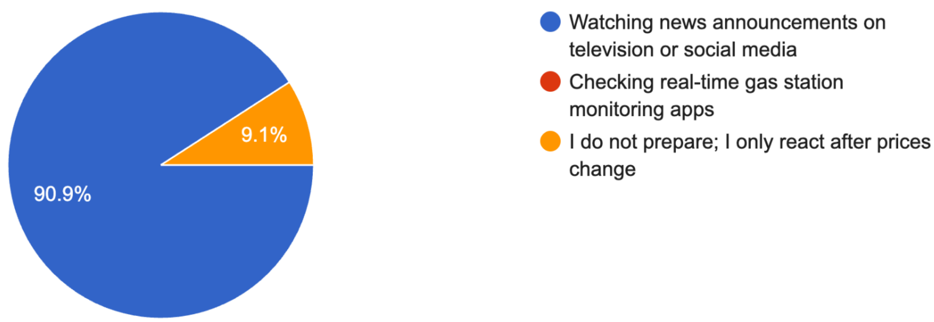
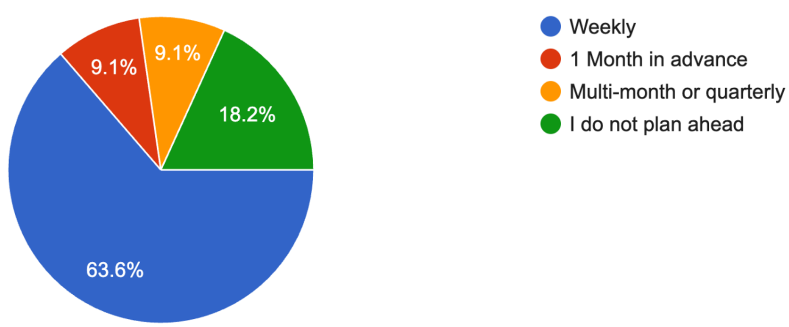


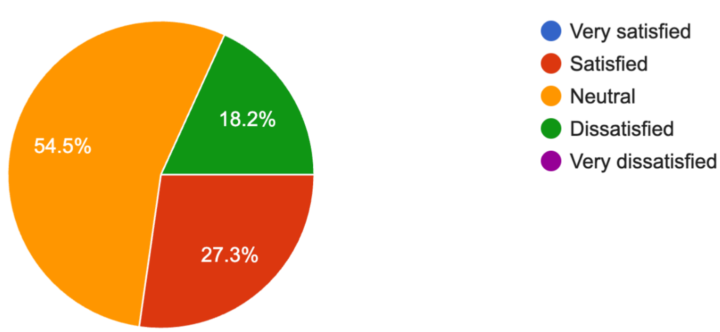
How far in advance do you attempt to plan your delivery costs or personal transport budgets?

11 responses



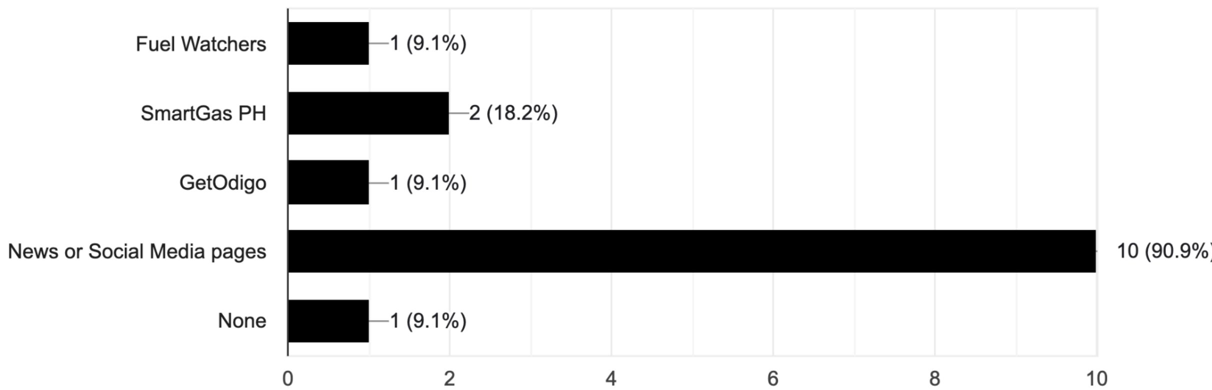
How satisfied are you with existing fuel tools that only display current or historical prices?

11 responses



Which of the following fuel tracking platforms have you used before? (Select all that apply)

11 responses



Which design aspect is most important to you when choosing a tracking app?

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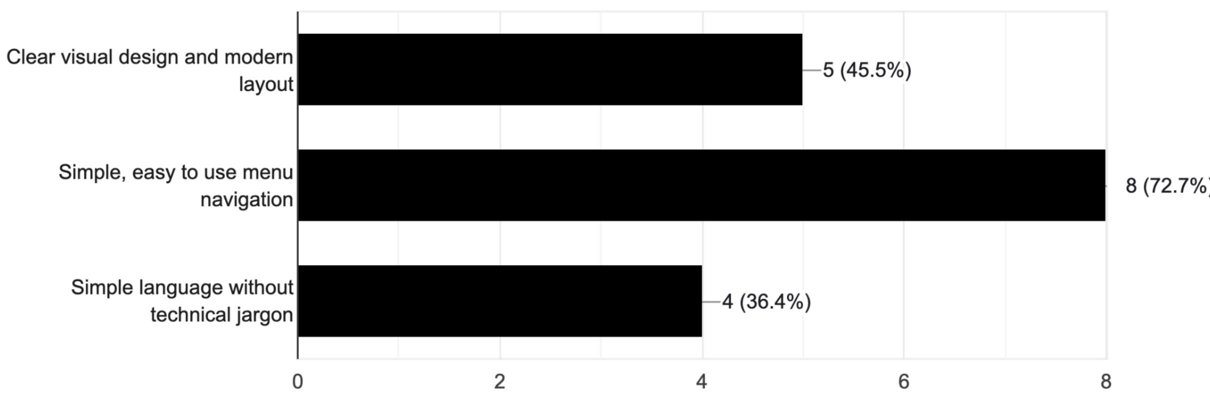

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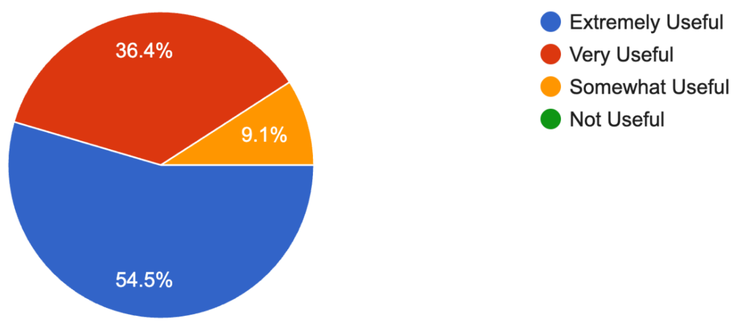

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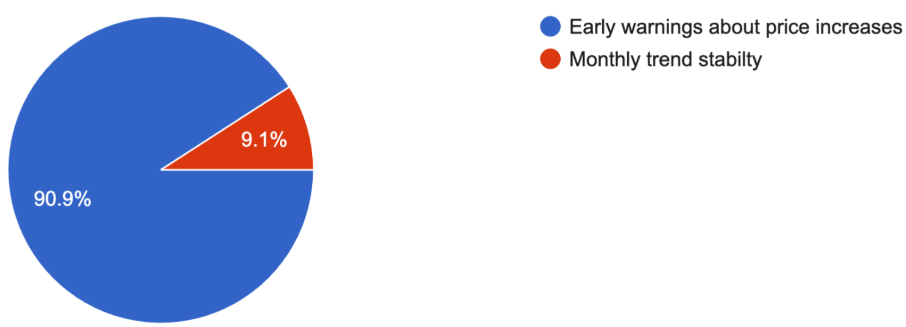
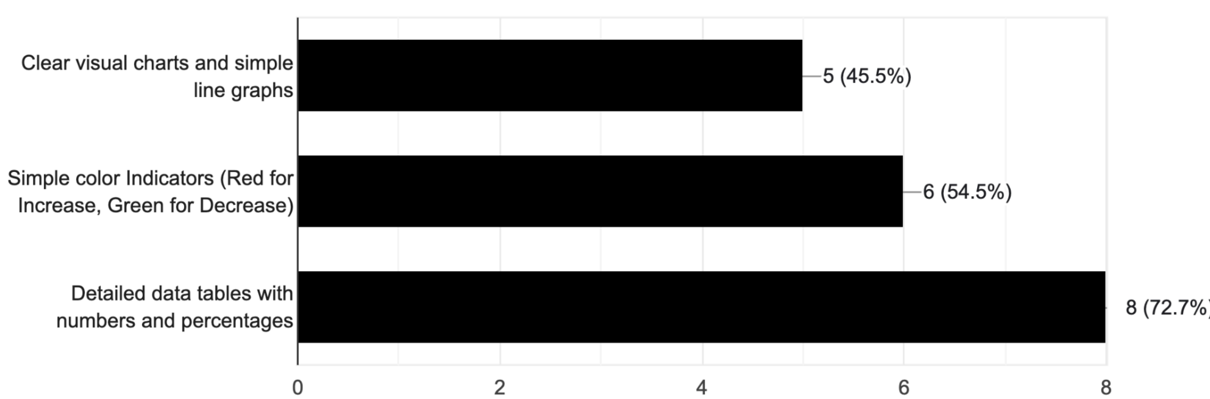


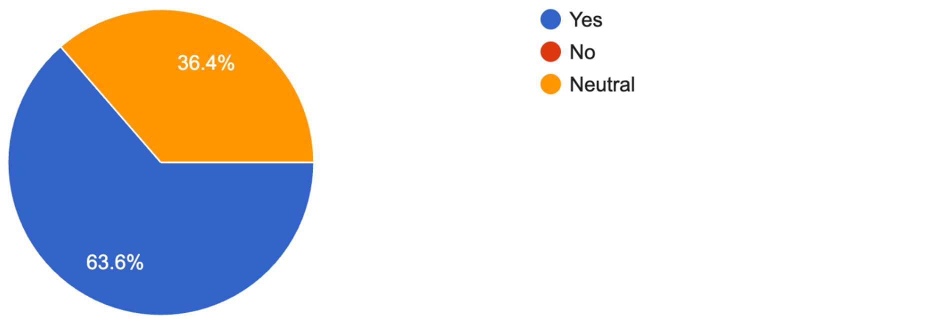
How would you like the 30-day forecast to be presented on your screen

11 responses



FuelCast uses automated systems combining shipping, market, and economic data to make predictions. Does this build your trust in the system?

11 responses



Which visual style makes it easier for you to make quick decisions?

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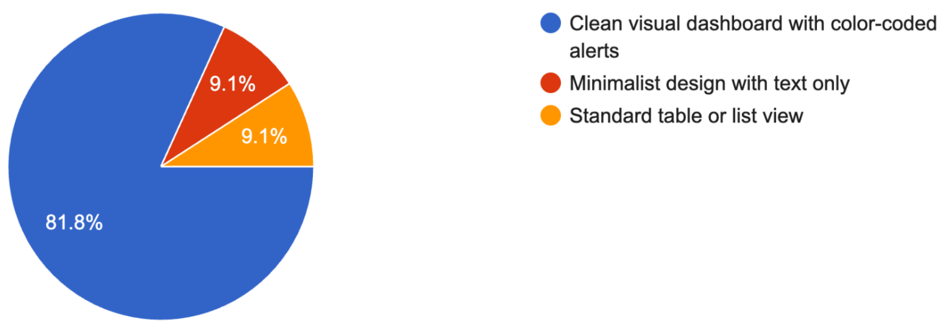

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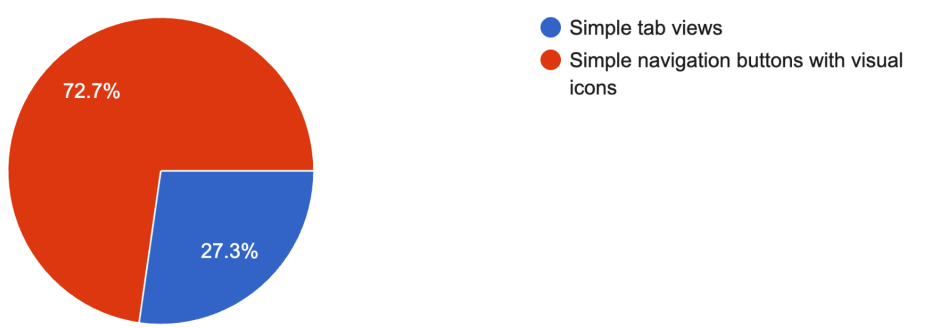

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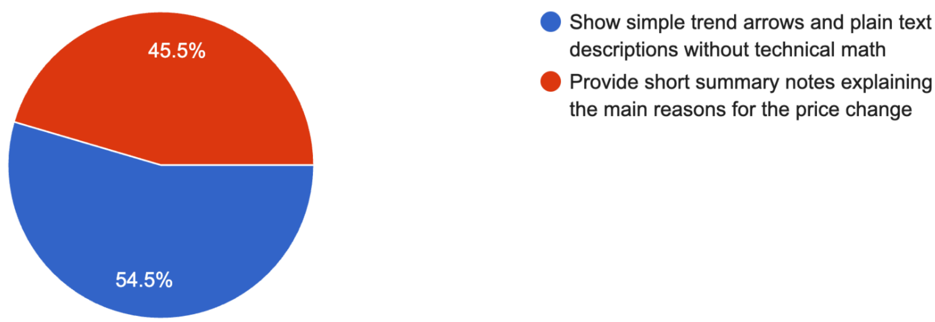
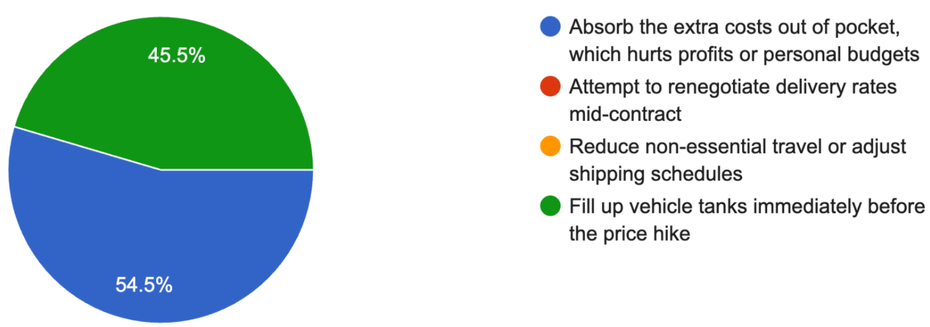


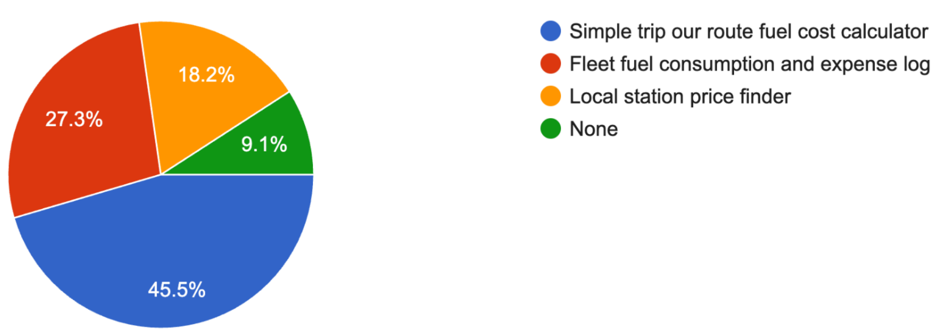
If fuel prices suddenly increase, what actions do you usually take?

11 responses



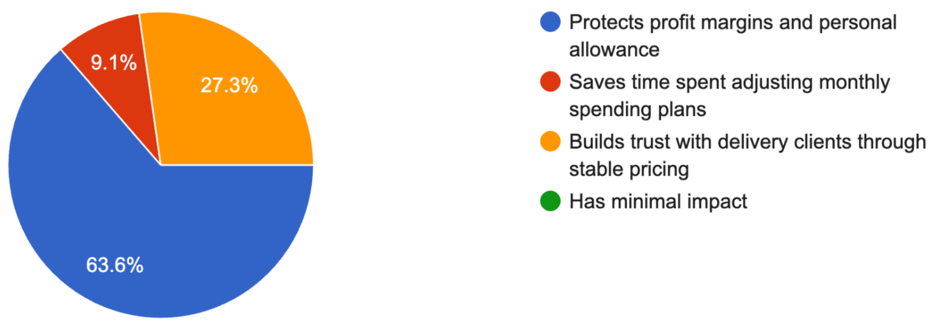
Which additional expense tool would be most helpful inside FuelCast

11 responses



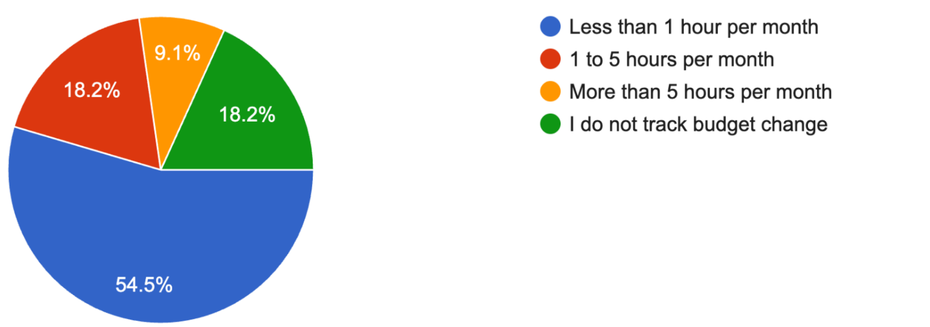
How would predictable fuel costs affect your operational overall?

11 responses



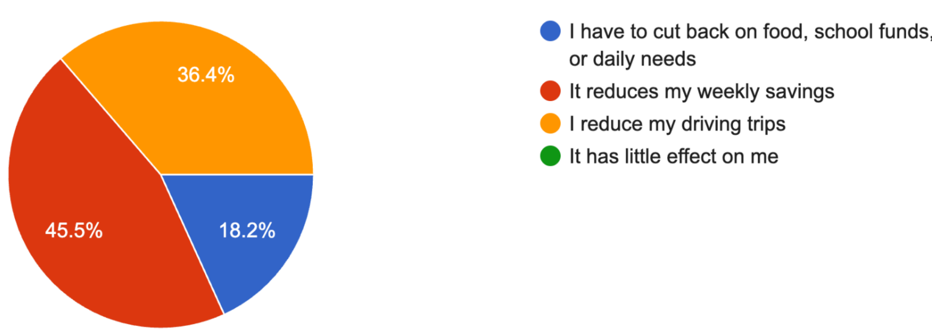
How much time do you currently spend adjusting budgets due to changing gas prices

11 responses



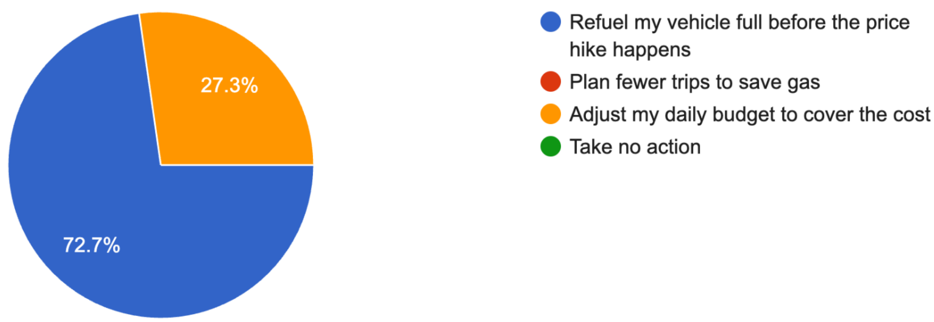
When gas prices spike unexpectedly, how does it affect your personal spending?

11 responses



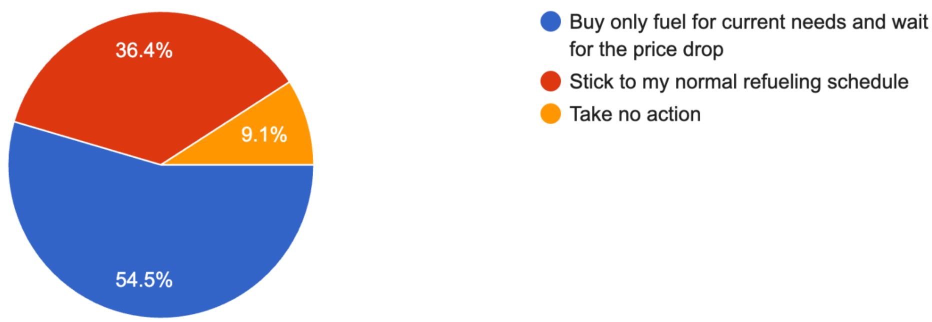
If FuelCast alerts you that gas prices will rise by PHP 2.00 next week, what will you do?

11 responses



If FuelCast alert you that gas prices will drop next week, what will you do?

11 responses



How far in advance do you need a price warning to adjust your personal routine?

11 responses